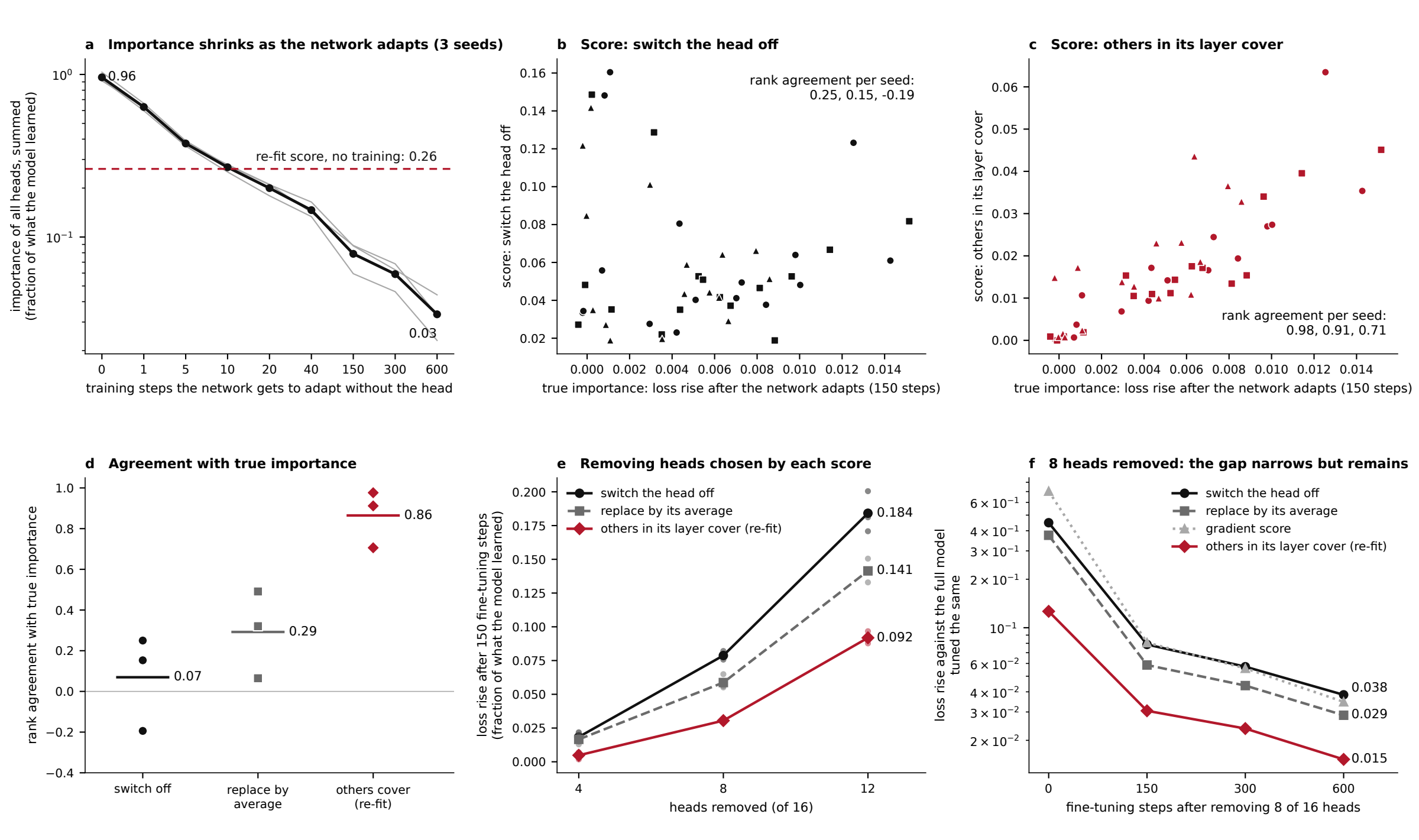
a Importance shrinks as the network adapts (3 seeds)
importance of all heads, summed (fraction of what the model learned)
0.96
re-fit score, no training: 0.26
0.03
training steps the network gets to adapt without the head
b Score: switch the head off
score: switch the head off
rank agreement per seed: 0.25, 0.15, -0.19
true importance: loss rise after the network adapts (150 steps)
c Score: others in its layer cover
score: others in its layer cover
rank agreement per seed: 0.98, 0.91, 0.71
true importance: loss rise after the network adapts (150 steps)
d Agreement with true importance
rank agreement with true importance
0.86
0.29
0.07
switch off
replace by average
others cover (re-fit)
e Removing heads chosen by each score
loss rise after 150 fine-tuning steps (fraction of what the model learned)
switch the head off
replace by its average
others in its layer cover (re-fit)
0.184
0.141
0.092
heads removed (of 16)
f 8 heads removed: the gap narrows but remains
loss rise against the full model tuned the same
switch the head off
replace by its average
gradient score
others in its layer cover (re-fit)
0.038
0.029
0.015
fine-tuning steps after removing 8 of 16 heads